

OB Q

Broadcast-desk dynamics - gate / expander, compressor, limiter and transformer - with a four-band channel EQ.



Overview

OB Q is the centre-section dynamics of an 80s British broadcast console, in the spirit of the classic desk VCA compressor and its companion compressor / gate module. It is fast, clean and transparent, with the console's output transformer for a little weight. The signal goes gate / expander, then compressor, then limiter, then transformer.

Below the dynamics sits a four-band console channel EQ: high- and low-pass filters, LF and HF bands (shelf or bell), and two fully parametric mid bands, with two voicings, BROWN and BLACK.

ORDER puts the EQ after the dynamics or in front of them. The unit the signal meets first is shown on top.

Reach for it for

- Vocals and voice-over that need to sit steady and clear

- Tight, gated drums (kick, snare, toms)
- Cleaning up noise and spill between phrases (EXP)
- A fast, transparent bus compressor with a touch of transformer weight
- Channel EQ and dynamics in one place, in either order

Controls

Gate / Expander

THRESH	The level below which the gate or expander works, -80 to 0 dB. Fully left = OFF.
RANGE	How far it closes, 0 - 60 dB. Small settings just lower the noise between phrases; 60 dB is fully shut.
RELEASE	How quickly it closes once the signal falls below the threshold, 20 - 2000 ms. It opens in about half a millisecond.
IN	Switches the gate / expander in or out (out by default).
GATE / EXP	GATE: shuts by the full RANGE as soon as the level drops below the threshold. EXP (default): a gentle 1:2 expander that only takes down what falls below - the further below, the quieter, up to RANGE.
OPEN	Lights while the gate is letting the signal through.
GATE readout	How many dB the gate is taking off, or GATE OUT when it is switched out.

Compressor

THRESH	Where the compressor starts working, -40 to 0 dB. It has a 3 dB soft knee and a quick RMS detector, so it holds the level without dulling the transients.
RATIO	1:1 to INF:1 (fully right).
ATTACK	0.1 - 30 ms.

RELEASE	30 - 3000 ms.
GAIN	Make-up gain, -10 to +20 dB.
IN	Switches the compressor in or out (in by default). GAIN only applies while it is in.
GR meter	The row of 12 lights shows the gain reduction of the compressor and limiter together, from 1 to 30 dB.

Limit / Output

LIMIT	A fast peak limiter after the compressor, -20 to 0 dB: the level the peaks are held at. Fully right = OFF. The red light beside it shows it limiting.
XFMR	The output transformer, on by default: the lows thicken and saturate gently as the level rises, and the extreme top softens. Off = straight wire.
LINK	On (default): both sides share one gain for the gate, compressor and limiter, so the stereo image stays put. Off: two independent mono channels.
MIX	Parallel compression, 0 - 100% wet: blends the dry signal back in. The dry signal is taken after the EQ when the EQ comes first.
IN / OUT meters	LED ladders for the input and output level; click one to clear the peak hold.
GR readout	The total gain reduction in dB.

EQ

The four-band channel EQ strip.

IN	Switches the four EQ bands in or out (in by default). The HPF and LPF keep working either way.
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BROWN / BLACK	The voicing. BROWN: broad, gentle curves whose width stays the same at any gain. BLACK (default): the bandwidth narrows as you boost or cut further, for a more focused sound, and the shelves get a small opposite dip just inside their corner.
HPF	High-pass filter, 18 dB/oct, 20 - 400 Hz. Fully left = OFF.
LPF	Low-pass filter, 12 dB/oct, 3 - 22 kHz. Fully right = OFF.
LF / FREQ	Low band: -15 to +15 dB at 30 - 450 Hz.
BELL (LF)	Off: a low shelf. On: a bell curve at FREQ.
LMF / FREQ / Q	Low-mid band: -15 to +15 dB, 200 Hz - 2.5 kHz, Q 0.4 - 3 (higher is narrower).
HMF / FREQ / Q	High-mid band: -15 to +15 dB, 600 Hz - 7 kHz, Q 0.4 - 3.
HF / FREQ	High band: -15 to +15 dB at 1.5 - 16 kHz.
BELL (HF)	Off: a high shelf. On: a bell curve at FREQ.

The header

ORDER	DYNAMICS > EQ (default) or EQ > DYNAMICS: click to swap. The panel redraws with the unit the signal meets first on top.
PRESETS	Opens the list of factory presets; the ◀ / ▶ keys step through them and the display shows the one in use (click it for the list too). Choosing one sets the gate, compressor, LIMIT, XFMR, LINK and MIX; the EQ is left as it is. The last entry resets the dynamics to their defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Input guard

There is no input or output fold-out on OB Q: its own character does that job. An input guard runs in front of everything and is on by default: it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources. It has no key on the panel; it appears as Input guard in REAPER's parameter list.

In REAPER's parameter list only

EQ: Line amp drive	Drives the EQ's line amplifier into gentle, slightly asymmetric saturation, 0 - 100% (0 by default).
EQ: Output	The EQ's output level, -12 to +12 dB.

Presets

Desk Glue	Gentle 2.5:1 compression with linked sides for a bus; gate out.
Presenter Voice	Expander for the room noise, 4:1 compression with 6 dB of make-up and a limit at -2 dB.
Tight Drums (gated)	Gate closing by 40 dB, fast 5:1 compression, a limit at -1 dB.
Kick / Snare Gate	A hard, fast gate closing fully, fast 4:1 compression, sides unlinked.
Parallel Squash	Heavy 8:1 compression with a fast attack, blended in at 45%.
Transmitter Limit	INF:1 with the fastest attack at -14 dB, then a limit at -1 dB.
Acoustic Clean-up	A gentle expander under 3:1 compression with slower timing.
Bass DI	Steady 4:1 compression with an 8 ms attack.
Transformer Only	Compressor and gate out: just the output transformer.

Gentle Programme Bus	A light expander and slow 2:1 compression, with a limit at -1 dB.
Reset all to defaults	The dynamics and the input guard back to their starting values; the EQ is left as it is.

Tips

- EXP is far more forgiving than GATE on vocals and acoustic instruments: set RANGE to 10 - 15 dB to lower the noise without chopping the tails.
- EQ > DYNAMICS lets an HPF stop the compressor reacting to rumble; DYNAMICS > EQ keeps your EQ boosts from changing how hard it compresses.
- Switch LINK off on drum overheads or a wide stereo source if you want each side to breathe on its own.
- Try BLACK for surgical mid cuts and BROWN for broad, musical tone shaping.

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